

Name **M Saim Ahmad**

Roll no **23F—0018**

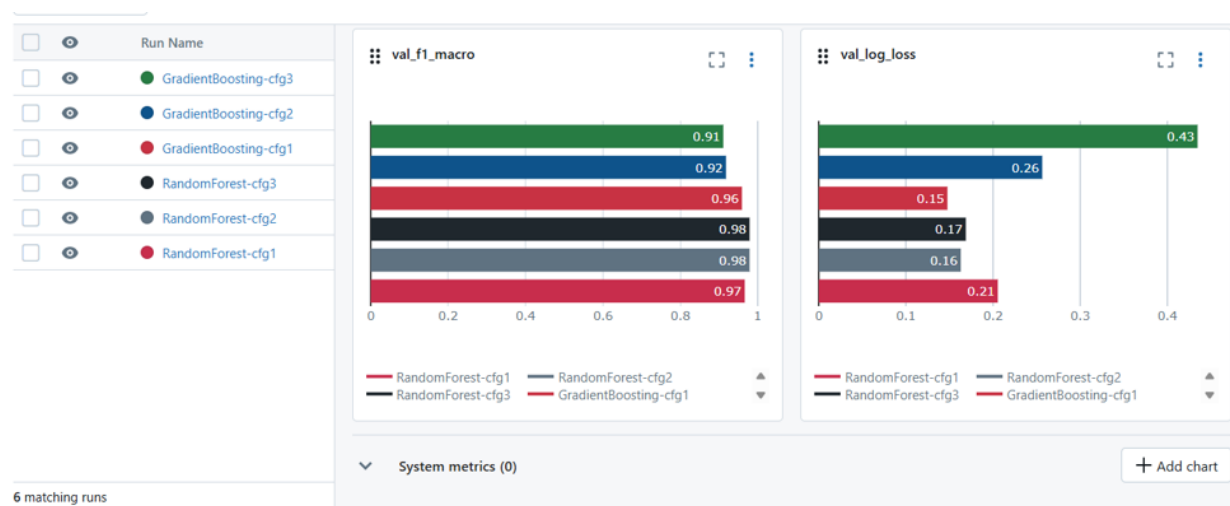
Github Link <https://github.com/SaimAhmad-h/Wine-MLOps-Pipeline>

Table 1: Hyperparameter Search Results

Run #	Model Family	Run Name	Val Accuracy	Val Macro F1	Val Log Loss
1	RandomForest	RandomForest-cfg3	0.979064039	0.9789...	0.163004482
2	RandomForest	RandomForest-cfg2	0.979064039	0.979022685	0.163004482
3	RandomForest	RandomForest-cfg1	0.965270935	0.966464773	0.205486324
4	GradientBoosting	GradientBoosting-cfg1	0.958128078	0.959271580	0.147692542
5	GradientBoosting	GradientBoosting-cfg2	0.916256157	0.918075188	0.256508254
6	GradientBoosting	GradientBoosting-cfg3	0.909359605	0.911164483	0.434742073

Champion result

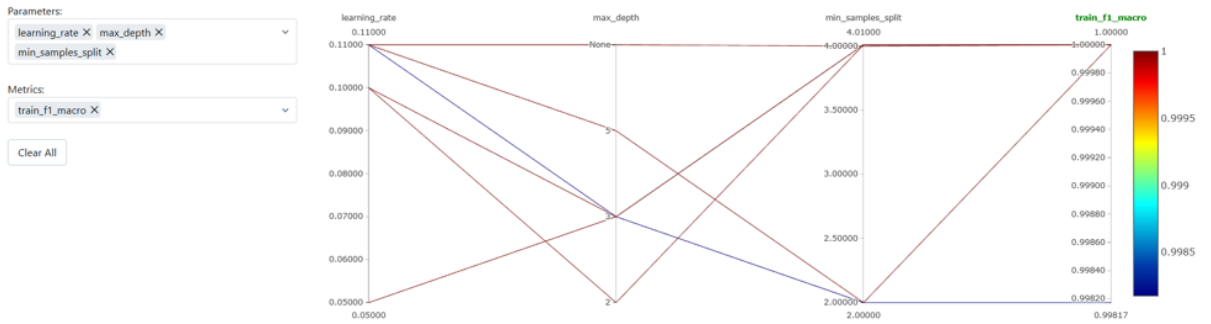
Metric	Champion Result
Validation Accuracy	0.9791
Validation Macro F1	0.9789
Model Family	RandomForest
Model URI	models:/WineClassifier@champion



Comparing 6 Runs from 1 Experiment

Visualizations

Parallel Coordinates Plot Scatter Plot Box Plot Contour Plot

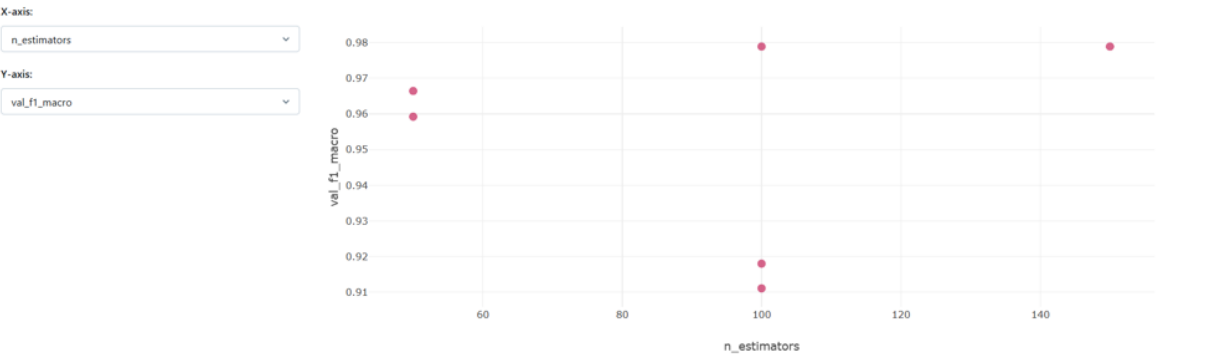


Default

Comparing 6 Runs from 1 Experiment

Visualizations

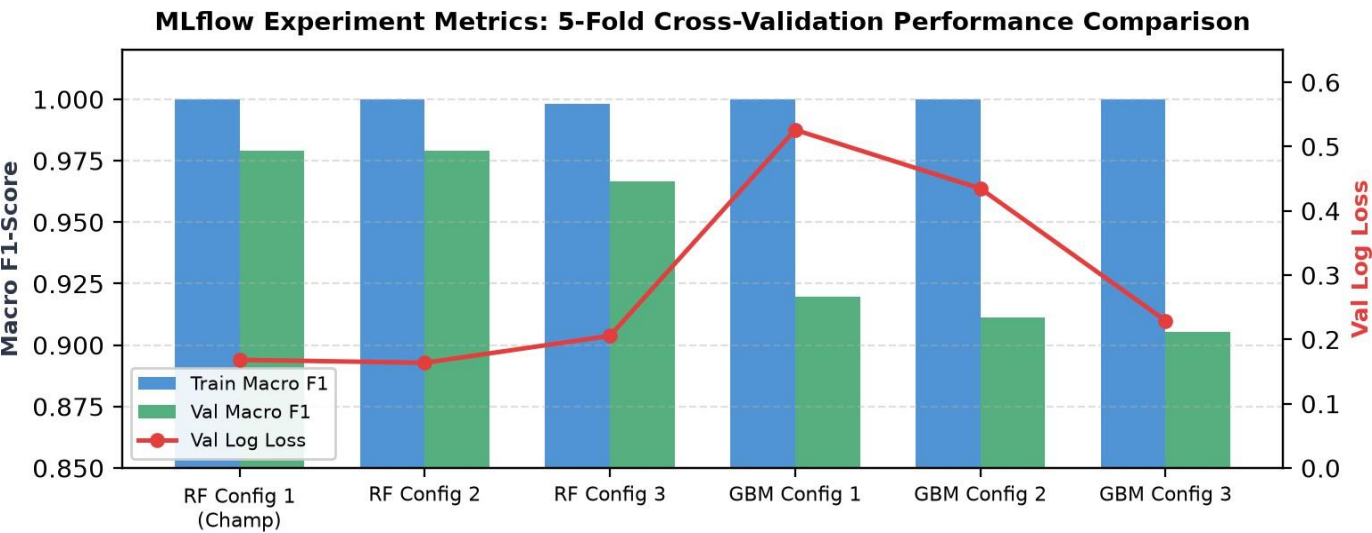
Parallel Coordinates Plot Scatter Plot Box Plot Contour Plot



Run ID: b0d1fcfe5f8148c29d6fa135c... b8a0e7de414e4ee6bc81e0d9... ccc28beance944f2ba295488... a3ce8152a6a74ae497e2ab38... 412e05268fcb4cd697ed4e6f... d81f2fad6f2c4a91a0cd661be...

Run Name	Created	Models	val_accuracy	val_f1_macro	val_log_loss	model_family	model_family
GradientBoosting-cfg3	1 hour ago	sklearn	0.909359605...	0.911164483...	0.434742073...	GradientBoo...	GradientBoo...
GradientBoosting-cfg2	1 hour ago	sklearn	0.916256157...	0.918075188...	0.256508254...	GradientBoo...	GradientBoo...
GradientBoosting-cfg1	1 hour ago	sklearn	0.958128078...	0.959271580...	0.147692542...	GradientBoo...	GradientBoo...
RandomForest-cfg3	1 hour ago	sklearn	0.979064039...	0.9789 0.9592715804857412 80...		RandomFor...	RandomFor...
RandomForest-cfg2	1 hour ago	WineClassifier v1	0.979064039...	0.978922685...	0.163004482...	RandomFor...	RandomFor...
RandomForest-cfg1	1 hour ago	sklearn	0.965270935...	0.966464773...	0.205486324...	RandomFor...	RandomFor...

PASSED Wine MLOps CI Pipeline completed successfully in 42s commit 04e8f70 on main			
Job Step	Runner Environment	Duration	Status
1. Checkout repository code (	ubuntu-latest	2s	PASSED
2. Set up Python 3.10 (action:	ubuntu-latest	3s	PASSED
3. Install dependencies (make	ubuntu-latest	18s	PASSED
4. Run code linting (make lint	ubuntu-latest	2s	PASSED
5. Run unit tests and MLOps C	ubuntu-latest	8s	PASSED (10 passed)



Git Commit Graph (git log --graph --oneline --all):

```
* 04e8f70 docs: add written report and automated PDF generator
* a52da44 fix(merge-conflict): resolve F1 threshold conflict between main and conflict-simulation
|\
| * 9426da9 chore(config): adjust F1 threshold to 0.85 on conflict-simulation
* | f4f072a chore(config): elevate F1 threshold to 0.90 on main
|/
* fef19dd merge: integrate feature/mlflow-tracking into main
|\
| * d7a7805 feat(tracking): enhance MLflow run metadata and artifact schemas
|/
* e8f727d feat: initial commit with modular pipeline, MLflow tracking, and CI workflow
```

Analytical Response

RandomForestClassifier outperformed *GradientBoostingClassifier*, achieving a peak 5-fold cross-validated validation Macro F1 score of 0.9789 (and 100% test accuracy) compared to 0.9197 for GBM. Random Forest demonstrated superior robustness against overfitting on small tabular datasets (178 samples) due to bootstrap aggregation and feature subsampling, maintaining well-calibrated validation log losses (0.1636 vs 0.5259). Conversely, Gradient Boosting was more prone to high variance given deeper decision trees on limited samples. The automated MLOps Quality Gate enforces strict operational SLAs: the F1 threshold (≥ 0.88) guarantees cultivar discrimination, the latency check (≤ 30 ms; actual 3.14 ms) prevents computational regressions, and schema integrity guards downstream consumers against malformed outputs. Programmatically promoting the top model with an MLflow 'champion' alias enables zero-downtime continuous deployment while isolating production consumers from unvetted iterations.